

- (1) Given two points $a, b \in \mathbb{R}$, with $a < b$, consider $f(x) = \max\{|x - a|, |x - b|\}$. Find
- (a) $f_* = \min_{x \in \mathbb{R}} f(x)$
 - (b) $x_* = \arg \min_{x \in \mathbb{R}} f(x)$
 - (c) Is the minimiser x_* unique?

Solve and explain this properly to me. Do not assume anything beforehand.

We're given real numbers $a, b \in \mathbb{R}$ with $a < b$, and

$$f(x) = \max\{|x - a|, |x - b|\}.$$

Interpretation: $f(x)$ is the larger of the two distances from x to the points a and b . We want the point x that makes the *worst* (largest) of these two distances as small as possible.

1) Understand $f(x)$ by splitting into cases

Because absolute values change form depending on where x lies relative to a and b , consider three regions.

Case A: $x \leq a$

Then $x - a \leq 0$ and $x - b \leq 0$, so

$$|x - a| = a - x, \quad |x - b| = b - x.$$

Since $b > a$, we have $b - x > a - x$. Therefore the max is always the bigger one:

$$f(x) = b - x \quad (x \leq a).$$

This is a decreasing linear function in x (as x increases, $b - x$ decreases).

Case B: $a \leq x \leq b$

Then $x - a \geq 0$ and $x - b \leq 0$, so

$$|x - a| = x - a, \quad |x - b| = b - x.$$

So

$$f(x) = \max\{x - a, b - x\} \quad (a \leq x \leq b).$$

Here one term increases with x ($x - a$), and the other decreases with x ($b - x$).

Case C: $x \geq b$

Then $x - a \geq 0$ and $x - b \geq 0$, so

$$|x - a| = x - a, \quad |x - b| = x - b.$$

Since $x - a > x - b$ (because $a < b$), the max is

$$f(x) = x - a \quad (x \geq b),$$

which is increasing linear in x .

2) Where can the minimum occur?

- For $x \leq a$: $f(x) = b - x$ decreases as x increases, so in that region the smallest value happens at the **largest** x , i.e. at $x = a$.

$$f(a) = \max\{0, b - a\} = b - a.$$

- For $x \geq b$: $f(x) = x - a$ increases as x increases, so in that region the smallest value happens at the **smallest** x , i.e. at $x = b$.

$$f(b) = \max\{b - a, 0\} = b - a.$$

So outside $[a, b]$, you can't do better than $b - a$, and you get that only at the endpoints a or b .

But inside $[a, b]$ we might do better. Let's minimize there.

3) Minimize on $[a, b]$

On $[a, b]$,

$$f(x) = \max\{x - a, b - x\}.$$

To make the maximum of two numbers as small as possible, you try to make them **equal** (otherwise the larger one can often be reduced by moving toward balancing them).

So set

$$x - a = b - x.$$

Solve carefully:

$$x - a = b - x$$

Add x to both sides:

$$2x - a = b$$

Add a :

$$2x = a + b$$

Divide by 2:

$$x = \frac{a + b}{2}.$$

At this point both distances are the same:

$$\left| \frac{a + b}{2} - a \right| = \frac{b - a}{2}, \quad \left| \frac{a + b}{2} - b \right| = \frac{b - a}{2}.$$

So

$$f\left(\frac{a + b}{2}\right) = \frac{b - a}{2}.$$

And $\frac{b-a}{2} < b - a$, so this is indeed better than anything achievable outside $[a, b]$.

Answers

(a) The minimum value f_*

$$f_* = \min_{x \in \mathbb{R}} f(x) = \frac{b - a}{2}.$$

(b) The minimizer x_*

$$x_* = \arg \min_{x \in \mathbb{R}} f(x) = \frac{a + b}{2}.$$

("argmin" here is the point where the minimum is attained.)

(c) Is the minimizer unique?

Yes.

Reason (without assuming convexity):

- On $[a, b]$, the function is $\max\{x - a, b - x\}$.
 - For $x < \frac{a+b}{2}$, we have $b - x > x - a$, so $f(x) = b - x$, which strictly decreases as x increases.
 - For $x > \frac{a+b}{2}$, we have $x - a > b - x$, so $f(x) = x - a$, which strictly increases as x increases.
- So it decreases up to $\frac{a+b}{2}$ and increases after, meaning there is exactly **one** point where it's smallest.

Yes, the minimizer is unique, and it is $x_* = \frac{a+b}{2}$.

If you want, I can also sketch the graph shape in words (it's a "V" with its bottom at the midpoint) or derive it using a geometric "smallest radius covering both points" argument.



(2) The per Kg cost of potato is Rs.20/–, tomato is Rs.25/– and Onion is Rs.30/–. We have Rs.100/– in hand we want to buy as much as possible. *We have to buy at least two different items.* Express this as an optimisation problem. (Hint: First decide the number of variables, the domain of each variable and number of constraints).

Let's build the optimization model from scratch, exactly as the hint asks: **variables** → **domains** → **constraints** → **objective**.

1) Decide the variables (how many?)

We are choosing how many kilograms of each item to buy:

- x_1 = kg of **potato**
- x_2 = kg of **tomato**
- x_3 = kg of **onion**

So we need **3 decision variables**.

2) Domain of each variable

"Kg to buy" can't be negative. Unless the question explicitly says we must buy whole kilograms, we should allow fractional kg.

So:

$$x_1, x_2, x_3 \in \mathbb{R}_{\geq 0}.$$

(If your instructor wants whole kgs, replace by $x_i \in \mathbb{Z}_{\geq 0}$. I'll write the continuous version first.)

3) Constraints (how many and what?)

(i) Budget constraint

Costs per kg:

- potato: 20
- tomato: 25
- onion: 30

We have Rs. 100, so:

$$20x_1 + 25x_2 + 30x_3 \leq 100.$$

(ii) "At least two different items"

This is not automatically enforced by $x_i \geq 0$. We must ensure **at least two of the three variables are strictly positive**.

To express "use at least two items" cleanly, we introduce **binary variables** that indicate whether an item is purchased.

Let:

$$y_i = \begin{cases} 1, & \text{if item } i \text{ is bought (positive amount)} \\ 0, & \text{otherwise} \end{cases} \quad i = 1, 2, 3$$

So $y_1, y_2, y_3 \in \{0, 1\}$.

Then enforce:

$$y_1 + y_2 + y_3 \geq 2.$$

Now we must link x_i and y_i so that:

- if $y_i = 0$, then $x_i = 0$
- if $y_i = 1$, then x_i is allowed to be positive

A standard way is:

$$0 \leq x_i \leq M_i y_i \quad i = 1, 2, 3$$

for some valid upper bounds M_i .

We can choose tight M_i using the budget: maximum kg of each item if you spent all 100 on it:

- Potato: $100/20 = 5 \Rightarrow M_1 = 5$
- Tomato: $100/25 = 4 \Rightarrow M_2 = 4$
- Onion: $100/30 = 10/3 \Rightarrow M_3 = 10/3$

So linking constraints:

$$0 \leq x_1 \leq 5y_1, \quad 0 \leq x_2 \leq 4y_2, \quad 0 \leq x_3 \leq \frac{10}{3}y_3.$$

That fully models "at least two different items".

4) Objective: "buy as much as possible"

"As much as possible" means maximize total kilograms purchased:

$$\max x_1 + x_2 + x_3.$$

Final optimization problem (MILP)

$$\begin{aligned} \max_{x_1, x_2, x_3, y_1, y_2, y_3} \quad & x_1 + x_2 + x_3 \\ \text{s.t.} \quad & 20x_1 + 25x_2 + 30x_3 \leq 100 \\ & 0 \leq x_1 \leq 5y_1 \\ & 0 \leq x_2 \leq 4y_2 \\ & 0 \leq x_3 \leq \frac{10}{3}y_3 \\ & y_1 + y_2 + y_3 \geq 2 \\ & x_1, x_2, x_3 \geq 0 \\ & y_1, y_2, y_3 \in \{0, 1\}. \end{aligned}$$

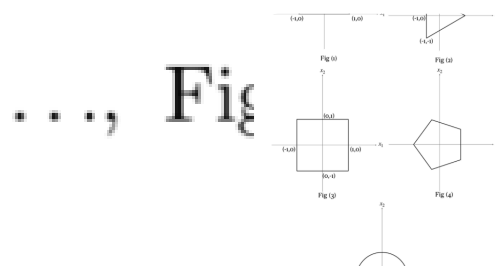
Counts (as the hint asked)

- **Variables:** 3 continuous (x_1, x_2, x_3) + 3 binary $(y_1, y_2, y_3) = 6$
- **Domains:** $x_i \geq 0, y_i \in \{0, 1\}$
- **Constraints:**
 - 1 budget constraint
 - 3 linking constraints $x_i \leq M_i y_i$
 - 1 "at least two items" constraint

Total = **5 constraints** (not counting nonnegativity and integrality conditions)

If you want, I can also show a version **without binary variables** (it's possible but awkward because "at least two are positive" is a non-convex logical condition).

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Let $(x_1, x_2) \in \mathbb{R}^2$. In each figure the **constraint set** is the **filled-in region including the boundary**.

Fig (1)

Triangle with vertices $(-1, 0)$, $(1, 0)$, $(0, 1)$.

The three edges are:

- base: $x_2 = 0$
- left side through $(-1, 0)$ and $(0, 1)$: $x_2 = x_1 + 1$
- right side through $(1, 0)$ and $(0, 1)$: $x_2 = -x_1 + 1$

So the feasible set is

$$\{(x_1, x_2) : x_2 \geq 0, x_2 \leq x_1 + 1, x_2 \leq -x_1 + 1\}$$

(equivalently $0 \leq x_2 \leq 1 - |x_1|$ and $|x_1| \leq 1$).

Fig (2)

Triangle with left vertical side from $(-1, -1)$ to $(-1, 1)$, and right apex at $(1, 0)$.

Edges:

- left side: $x_1 = -1$
- top edge through $(-1, 1)$ and $(1, 0)$: slope $-\frac{1}{2}$, so $x_2 = \frac{1-x_1}{2}$
- bottom edge through $(-1, -1)$ and $(1, 0)$: slope $+\frac{1}{2}$, so $x_2 = \frac{x_1-1}{2}$

Interior is to the right of $x_1 = -1$, and between the two slanted lines:

$$\{(x_1, x_2) : x_1 \geq -1, \frac{x_1 - 1}{2} \leq x_2 \leq \frac{1 - x_1}{2}\}$$

Fig (3)

Square with vertices $(\pm 1, \pm 1)$. Hence

$$\{(x_1, x_2) : -1 \leq x_1 \leq 1, -1 \leq x_2 \leq 1\}$$

(equivalently $|x_1| \leq 1, |x_2| \leq 1$).

Fig (4) (unit length for each segment)

This is a **regular pentagon** centered at the origin with **side length 1**, oriented so the **rightmost side is vertical**.

A clean algebraic description is as the intersection of 5 half-spaces:

$$\{x \in \mathbb{R}^2 : A_i^\top x \leq b, i = 1, \dots, 5\}$$

where $b = \frac{1}{2} \cot 36^\circ$, and the five normals are

$$A_1 = \begin{pmatrix} \sin 18^\circ \\ \cos 18^\circ \end{pmatrix}, \quad A_2 = \begin{pmatrix} \cos 36^\circ \\ -\sin 36^\circ \end{pmatrix}, \quad A_3 = \begin{pmatrix} \cos 36^\circ \\ \sin 36^\circ \end{pmatrix},$$

$$A_4 = \begin{pmatrix} \sin 18^\circ \\ -\cos 18^\circ \end{pmatrix}, \quad A_5 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

So explicitly:

$$\begin{aligned} \sin 18^\circ x_1 + \cos 18^\circ x_2 &\leq \frac{1}{2} \cot 36^\circ \\ \cos 36^\circ x_1 - \sin 36^\circ x_2 &\leq \frac{1}{2} \cot 36^\circ \\ \cos 36^\circ x_1 + \sin 36^\circ x_2 &\leq \frac{1}{2} \cot 36^\circ \\ \sin 18^\circ x_1 - \cos 18^\circ x_2 &\leq \frac{1}{2} \cot 36^\circ \\ x_1 &\leq \frac{1}{2} \cot 36^\circ \end{aligned}$$

(Numerically, $\frac{1}{2} \cot 36^\circ \approx 0.68819$.)

Fig (5)

Circle centered at the origin (unit circle). The filled disk is

$$\{(x_1, x_2) : x_1^2 + x_2^2 \leq 1\}$$

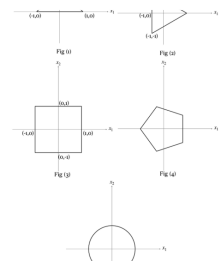
If you want, I can also give Fig (4) as “convex hull of vertices” with explicit vertex coordinates (using $\sin$, $\cos$ of multiples of 36°).



owing functions

$$\begin{aligned} &+x_2 \\ &1+x_2 \\ &-x_2 \\ &1-x_2 \\ &+x_2 \\ &-1)^2 + (x_2-2)^2 \\ &+3)^2 + (x_2+2)^2 \\ &1-1)^2 + 2(x_2-2)^2 \\ &+2x_2^2 + x_1 - x_2 + 4 \\ &+x_2^2 + 2x_1x_2 \end{aligned}$$

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and maximiser(s) for each of the constraint sets in Fig
ser unique?
and minimiser(s) for each of the constraint sets in Fig
unique?



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Below I'll treat each constraint set S_k (Fig 1–5) and, for each function f_i , give:

- **max value** and **maximiser(s)** on S_k ,

- **min value** and **minimiser(s)** on S_k ,
- whether the optimiser is **unique**.

I'll write points as (x_1, x_2) .

Constraint sets (from the figures)

Fig (1) (triangle): vertices

$$A = (-1, 0), \quad B = (1, 0), \quad C = (0, 1).$$

Fig (2) (triangle): vertices

$$D = (-1, 1), \quad E = (-1, -1), \quad F = (1, 0).$$

Fig (3) (square): vertices

$$(-1, -1), (-1, 1), (1, 1), (1, -1).$$

Fig (4) (regular pentagon, side length 1, centered at origin, right side vertical): vertices (exact trig form)

$$R = \frac{1}{2 \sin 36^\circ} \approx 0.8506508,$$

angles $36^\circ, -36^\circ, -108^\circ, -180^\circ, 108^\circ$, so

$$\begin{aligned} P_1 &= (0.68819096, 0.5), \\ P_2 &= (0.68819096, -0.5), \\ P_3 &= (-0.26286556, -0.80901699), \\ P_4 &= (-0.85065081, 0), \\ P_5 &= (-0.26286556, 0.80901699). \end{aligned}$$

Fig (5) (unit disk):

$$S_5 = \{x : x_1^2 + x_2^2 \leq 1\}.$$

The functions

Linear:

1. $x_1 + x_2$
2. $-x_1 + x_2$

3. $x_1 - x_2$

4. $-x_1 - x_2$

5. $2x_1 + x_2$

Quadratic:

6) $(x_1 - 1)^2 + (x_2 - 2)^2$

7) $(x_1 + 3)^2 + (x_2 + 2)^2$

8) $5(x_1 - 1)^2 + 2(x_2 - 2)^2$

9) $x_1^2 + 2x_2^2 + x_1 - x_2 + 4$

10) $x_1^2 + x_2^2 + 2x_1x_2 = (x_1 + x_2)^2$

Fig (1) results (triangle with A,B,C)

Linear (1)–(5)

1. $x_1 + x_2$:

- $\max = 1$, maximisers: entire edge $\overline{BC}$ (between $(1, 0)$ and $(0, 1)$) → not unique
- $\min = -1$ at $A = (-1, 0)$ → unique

2. $-x_1 + x_2$:

- $\max = 1$ at $A = (-1, 0)$ and $C = (0, 1)$ (edge $\overline{AC}$) → not unique
- $\min = -1$ at $B = (1, 0)$ → unique

3. $x_1 - x_2$:

- $\max = 1$ at $B = (1, 0)$ → unique
- $\min = -1$ at $A = (-1, 0)$ and $C = (0, 1)$ (edge $\overline{AC}$) → not unique

4. $-x_1 - x_2$:

- $\max = 1$ at $A = (-1, 0)$ → unique
- $\min = -1$, maximisers of negative = minimisers here: entire edge $\overline{BC}$ → not unique

5. $2x_1 + x_2$:

- $\max = 2$ at $B = (1, 0)$ → unique
- $\min = -2$ at $A = (-1, 0)$ → unique

Quadratic (6)–(10)

6. $(x_1 - 1)^2 + (x_2 - 2)^2$:

- $\min = 2$ at $C = (0, 1)$ → unique
- $\max = 8$ at $A = (-1, 0)$ → unique

7. $(x_1 + 3)^2 + (x_2 + 2)^2$:

- $\min = 8$ at $A = (-1, 0)$ → unique
- $\max = 20$ at $B = (1, 0)$ → unique

8. $5(x_1 - 1)^2 + 2(x_2 - 2)^2$:
 - $\min = 17$ at $C = (0, 1) \rightarrow$ **unique**
 - $\max = 52$ at $A = (-1, 0) \rightarrow$ **unique**
 9. $x_1^2 + 2x_2^2 + x_1 - x_2 + 4$:

Unconstrained minimiser is $(-1/2, 1/4)$, which lies inside the triangle.

 - $\min = 29/8 = 3.625$ at $(-\frac{1}{2}, \frac{1}{4}) \rightarrow$ **unique**
 - $\max = 6$ at $B = (1, 0) \rightarrow$ **unique**
 10. $(x_1 + x_2)^2$:
 - $\min = 0$ for all points on $x_2 = -x_1$ inside the triangle, i.e. segment $\{(t, -t) : -1 \leq t \leq 0\} \rightarrow$ **not unique**
 - $\max = 1$ for all points with $|x_1 + x_2| = 1$ inside the triangle: edge $\overline{BC}$ (where $x_1 + x_2 = 1$) and the vertex A (where $x_1 + x_2 = -1$) $\rightarrow$ **not unique**
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Fig (2) results (triangle D,E,F)

Linear (1)–(5)

1. $x_1 + x_2$:
 - $\max = 1$ at $F = (1, 0)$ and $D = (-1, 1)$ (edge $\overline{DF}$) $\rightarrow$ **not unique**
 - $\min = -2$ at $E = (-1, -1) \rightarrow$ **unique**
2. $-x_1 + x_2$:
 - $\max = 2$ at $D = (-1, 1) \rightarrow$ **unique**
 - $\min = -1$ at $F = (1, 0) \rightarrow$ **unique**
3. $x_1 - x_2$:
 - $\max = 2$ at $F = (1, 0)$?? (check: $1 - 0 = 1$; at E : $-1 - (-1) = 0$; at D : $-1 - 1 = -2$)
So $\max = 1$ at $F \rightarrow$ **unique**
 - $\min = -2$ at $D \rightarrow$ **unique**
4. $-x_1 - x_2$:
 - $\max = 2$ at $E = (-1, -1) \rightarrow$ **unique**
 - $\min = -1$ at F and D (edge $\overline{DF}$) $\rightarrow$ **not unique**
5. $2x_1 + x_2$:
 - $\max = 2$ at $F = (1, 0) \rightarrow$ **unique**
 - $\min = -3$ at $E = (-1, -1) \rightarrow$ **unique**

Quadratic (6)–(10) (evaluated over the triangle)

6. $\min = 4$ at F ; $\max = 13$ at E . (both unique)

7. $\min = 5$ at E ; $\max = 17$ at F . (both **unique**)
 8. $\min = 25$ at F ; $\max = 65$ at E . (both **unique**)
 9. $\min = 29/8$ at $(-1/2, 1/4)$ (inside) **unique**; $\max = 7$ at E **unique**
 10. $\min = 0$ on segment of intersection with $x_1 + x_2 = 0$ (inside this triangle it includes the point $D = (-1, 1)$ and part of the interior) $\rightarrow$ **not unique**; $\max = 4$ at $E = (-1, -1)$ **unique**
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Fig (3) results (square)

Linear (1)–(5)

1. $x_1 + x_2$: $\max = 2$ at $(1, 1)$ **unique**; $\min = -2$ at $(-1, -1)$ **unique**
2. $-x_1 + x_2$: $\max = 2$ at $(-1, 1)$ **unique**; $\min = -2$ at $(1, -1)$ **unique**
3. $x_1 - x_2$: $\max = 2$ at $(1, -1)$ **unique**; $\min = -2$ at $(-1, 1)$ **unique**
4. $-x_1 - x_2$: $\max = 2$ at $(-1, -1)$ **unique**; $\min = -2$ at $(1, 1)$ **unique**
5. $2x_1 + x_2$: $\max = 3$ at $(1, 1)$ **unique**; $\min = -3$ at $(-1, -1)$ **unique**

Quadratic (6)–(10)

6. $\min = 1$ at $(1, 1)$ **unique**; $\max = 13$ at $(-1, -1)$ **unique**
 7. $\min = 5$ at $(-1, -1)$ **unique**; $\max = 25$ at $(1, 1)$ **unique**
 8. $\min = 2$ at $(1, 1)$ **unique**; $\max = 65$ at $(-1, -1)$ **unique**
 9. $\min = 29/8$ at $(-1/2, 1/4)$ (inside) **unique**; $\max = 10$ at $(1, -1)$ **unique**
 10. $\min = 0$ on the whole diagonal segment $\{(t, -t) : -1 \leq t \leq 1\} \rightarrow$ **not unique**; $\max = 4$ at $(1, 1)$ and $(-1, -1) \rightarrow$ **not unique**
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Fig (4) results (regular pentagon)

For **linear** functions, maxima/minima occur at vertices (sometimes along an edge if tied).

Linear (1)–(5) (numerical points)

1. $x_1 + x_2$:
 - $\max$ at $P_1 = (0.68819, 0.5)$ **unique** (value ≈ 1.18819)
 - $\min$ at $P_3 = (-0.26287, -0.80902)$ **unique** (value ≈ -1.07188)
2. $-x_1 + x_2$:
 - $\max$ at $P_5 = (-0.26287, 0.80902)$ **unique** (value ≈ 1.07188)
 - $\min$ at $P_2 = (0.68819, -0.5)$ **unique** (value ≈ -1.18819)
3. $x_1 - x_2$:

- max at $P_2 = (0.68819, -0.5)$ **unique** (value ≈ 1.18819)
- min at $P_5 = (-0.26287, 0.80902)$ **unique** (value ≈ -1.07188)
- 4. $-x_1 - x_2$:
 - max at $P_3 = (-0.26287, -0.80902)$ **unique** (value ≈ 1.07188)
 - min at $P_1 = (0.68819, 0.5)$ **unique** (value ≈ -1.18819)
- 5. $2x_1 + x_2$:
 - max at $P_1 = (0.68819, 0.5)$ **unique** (value ≈ 1.87638)
 - min at $P_4 = (-0.85065, 0)$ **unique** (value ≈ -1.70130)

Quadratic (6)–(10) (min may be on an edge; computed accurately)

- 6. min at P_1 with value ≈ 2.068 (unique); max at P_3 value ≈ 8.962 (unique)
- 7. min at P_4 value ≈ 5.321 (unique); max at P_1 value ≈ 20.197 (unique)
- 8. min at P_1 value ≈ 17.312 (unique); max at P_3 value ≈ 54.213 (unique)
- 9. min at $(-1/2, 1/4)$ (it lies inside the pentagon) value $= 29/8$ (unique); max at P_2 value ≈ 7.041 (unique)
- 10. min $= 0$ along intersection of pentagon with line $x_2 = -x_1$ (a segment) $\rightarrow$ not unique; max at P_1 value ≈ 1.4118 (unique)

Fig (5) results (unit disk $x_1^2 + x_2^2 \leq 1$)

Linear functions (1)–(5)

For $f(x) = c^\top x$ with $c \neq 0$:

- max at $x = \frac{c}{\|c\|}$, value $= \|c\|$ (unique),
- min at $x = -\frac{c}{\|c\|}$, value $= -\|c\|$ (unique).

So:

- 1. $c = (1, 1)$: max at $(1/\sqrt{2}, 1/\sqrt{2})$, value $\sqrt{2}$; min at $(-1/\sqrt{2}, -1/\sqrt{2})$, value $-\sqrt{2}$.
- 2. $c = (-1, 1)$: max at $(-1/\sqrt{2}, 1/\sqrt{2})$, value $\sqrt{2}$; min opposite.
- 3. $c = (1, -1)$: max at $(1/\sqrt{2}, -1/\sqrt{2})$, value $\sqrt{2}$; min opposite.
- 4. $c = (-1, -1)$: max at $(-1/\sqrt{2}, -1/\sqrt{2})$, value $\sqrt{2}$; min opposite.
- 5. $c = (2, 1)$: max at $(2/\sqrt{5}, 1/\sqrt{5})$, value $\sqrt{5}$; min opposite.

Quadratics (6)–(10)

- 6. $(x - (1, 2))^2$:
Let $u = (1, 2)$, $\|u\| = \sqrt{5}$.

- min at $x = u/\|u\| = (1/\sqrt{5}, 2/\sqrt{5})$, value $(\sqrt{5} - 1)^2$ (unique)
 - max at $x = -u/\|u\| = (-1/\sqrt{5}, -2/\sqrt{5})$, value $(\sqrt{5} + 1)^2$ (unique)
7. $(x - (-3, -2))^2$: $\|(-3, -2)\| = \sqrt{13}$
- min at $(-3/\sqrt{13}, -2/\sqrt{13})$, value $(\sqrt{13} - 1)^2$ unique
 - max at $(3/\sqrt{13}, 2/\sqrt{13})$, value $(\sqrt{13} + 1)^2$ unique
8. $5(x_1 - 1)^2 + 2(x_2 - 2)^2$:
- min occurs on boundary at approximately $(0.61815, 0.78606)$, value ≈ 3.67635 (unique)
 - max on boundary at approximately $(-0.88640, -0.46292)$, value ≈ 29.92448 (unique)
9. $x_1^2 + 2x_2^2 + x_1 - x_2 + 4$:
Unconstrained minimiser $(-1/2, 1/4)$ lies inside the disk.
- min = $29/8$ at $(-1/2, 1/4)$ (unique)
 - max on boundary at approximately $(0.32699, -0.94503)$, value ≈ 7.16510 (unique)
10. $(x_1 + x_2)^2$:
- min = 0 for all points in the disk with $x_1 + x_2 = 0$ (a diameter) $\rightarrow$ not unique
 - max = 2 attained at $(1/\sqrt{2}, 1/\sqrt{2})$ and $(-1/\sqrt{2}, -1/\sqrt{2}) \rightarrow$ not unique

If you want, I can also show **how** to get each linear optimum quickly using the “move in direction of the gradient” rule, and how the quadratic minima relate to “closest point / projection” and level sets.

